

BMAT201L- Complex Variables and Linear Algebra-Question Bank



MODULE 1: Analytic Functions

1. Discuss the analyticity of the function $f(z) = \log z$ at the origin.

Answer: $f(z) = \log z$ is analytic function except at the origin.

2. Show that the function
$$f(z) = \begin{cases} \frac{xy^2(x+iy)}{x^2+y^4}, & \text{if } z \neq 0 \\ 0, & \text{if } z = 0 \end{cases}$$

is not analytic at origin, although C-R equations are satisfied at the origin.

Hint: Along the path $x = my^2$, $f(z)$ not differentiable at the origin

3. Show that the function $f(z) = \sqrt{|xy|}$ is not analytic at the origin, even though C-R equations are satisfied at the origin.

Hint: Along the path $y = mx$, $f(z)$ not differentiable at the origin

4. Prove that the functions $u = x^2 - y^2$ and $v = \frac{-y}{x^2 + y^2}$ are harmonic but $f(z) = u + iv$ is not analytic.
5. If $w = f(z) = x^2 + ay^2 - 2xy + i(bx^2 - y^2 + 2xy)$ is analytic, then find the values of a and b .

Answer: $a = -1$ and $b = 1$.

6. If $f(z)$ is analytic function, show that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$.

7. Find the analytic function $f(z) = u + iv$ whose imaginary part is $v(x, y) = \frac{2 \sin x \sin y}{\cos 2x + \cosh 2y}$ and hence find $u(x, y)$.

Answer: $f(z) = \sec z + c$ where c is a complex constant and hence $u(x, y) = \frac{\cos x \cosh y}{\cos^2 x + \sinh^2 y} + k$.

8. Find the analytic function $f(z) = u + iv$ whose real part is $u(x, y) = x^3 - 3xy^2$ and hence find $u(x, y)$.

Answer: $f(z) = ze^z + c$ and $v(x, y) = 3x^2y - y^3 + k$

9. If $f(z) = u + iv$ is an analytic function of z and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .

Answer: $f(z) = e^z + c$

10. If $f(z) = u + iv$ is an analytic function such that $u + v = x^2 - y^2 + 2xy$, find $f(z)$ in terms of z .

Answer: $f(z) = z^2 + c$

11. In a two dimensional flow of a fluid, the velocity function is $\phi(x, y) = -2xy + \frac{y}{x^2 + y^2}$, find the stream function $\psi(x, y)$.

Answer: $\psi(x, y) = x^2 - y^2 + \frac{x}{x^2 + y^2} + k$.

12. If $f(z) = u + iv$ represents the complex potential for an electric field and $u(x, y) = e^{2x}(x \cos 2y - y \sin 2y)$, find $v(x, y)$.

Answer: $f(z) = ze^{2z} + c$ and $v(x, y) = e^{2x}(x \sin 2y + y \cos 2y) + k$.

MODULE 2: Conformal and Bilinear Transformations

13. Find the image of the triangle whose vertices are $(0, 0)$, $(2, 0)$ and $(0, 2)$ in the z -plane under the transformation $w = z + 3 - 4i$.

Answer: The image is the triangle whose vertices are $(3, -2)$, $(3, -4)$ and $(5, -4)$. in the w -plane.

14. Find the image of the region $y > \frac{1}{2}$ in the z -plane under the transformation $w = (1 - i)z$.

Answer: $u + v > 1$.

15. Find the image of the region $|z - 2i| = 2$ in the z -plane under the transformation $w = \frac{1}{z}$.

Answer: $v = -\frac{1}{4}$.

16. Find the image of the region $1 < x < 2$ in the z -plane under the transformation $w = \frac{1}{z}$.

Answer: The infinite strip $1 < x < 2$ transformed into the region in between the circles $\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}$ and $\left(u - \frac{1}{4}\right)^2 + v^2 = \frac{1}{16}$ in the w -plane.

17. Find the image of the region bounded by the lines $x=1$, $y=1$ and $x+y=1$ in the z -plane under the transformation $w = z^2$.

Answer: The image of the rectangular region bounded by the lines $x=1$, $y=1$ and $x+y=1$ in the z -plane is the region in the w -plane which is bounded the parabolas

$$v^2 = 4(1-u), v^2 = 4(1+u) \text{ and } u^2 = 2\left(\frac{1}{2} - v\right).$$

18. Find the image of the y -axis under the transformation $w = e^z$.

Answer: Find the image of the y -axis (i.e., $x=0$) is the unit circle $u^2 + v^2 = 1$.

19. Under the transformation $w = z^2$ find the image of $|z-1|=1$.

Answer: Taking $z = re^{i\theta}$ and $w = Re^{i\phi}$. Then the image of $|z-1|=1$ in the z -plane is $R = 2(1 + \cos \phi)$ in the w -plane.

20. Determine the bilinear transformation that maps the points $1-2i$, $2+i$, $2+3i$ in z -plane into the points $2+i$, $1+3i$, 4 respectively in w -plane.

Answer: $w = \frac{z(10+3i) - (7+16i)}{4z - 7(1+i)}$

21. Determine the bilinear transformation that maps the points ∞ , i , 0 in z -plane into the points -1 , $-i$, 1 respectively in w -plane and under this transformation find the image of $|z| < 1$.

Answer: $w = \frac{1-z}{1+z}$ and the image of the circle $|z| < 1$ in the z -plane is $u > 0$ in the w -plane.

22. Find the invariant points of the bilinear transformation $w = \frac{2i-6z}{iz-3}$.

Hint: The roots of the bilinear transformation $z = \frac{2i-6z}{iz-3}$ are the invariant points or fixed points.

Answer: $z = i, 2i$.

MODULE 3: Complex Integration

23. Obtain the Taylor's series expansion of the function $f(z) = \frac{1}{z^2 - z - 6}$ about $z = 1$.

Answer: $f(z) = \frac{-1}{6} + \frac{1}{5} \left(\frac{1}{2^2} - \frac{1}{3^2} \right) (z-1) - \left(\frac{1}{2^3} + \frac{1}{3^3} \right) (z-1)^2 + \left(\frac{1}{2^4} - \frac{1}{3^4} \right) (z-1)^3 + \dots$

24. Obtain the Laurent series expansion of the function $f(z) = \frac{z}{z^2 - 4z + 3}$ in the region

$$1 < |z| < 3.$$

Answer: $f(z) = \frac{-1}{6} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \dots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right)$

25. Obtain the Laurent series of the function $f(z) = \frac{z}{(z+1)(z+3)}$ about $z = -2$.

Answer: $f(z) = \frac{2}{(z+2)} + 1 + (z+2) + (z+2)^2 + \dots$

26. Find all the Laurent series of the function $f(z) = \frac{7z-2}{z(z+1)(z-2)}$ about $z = -1$.

Answer: Laurent series of $f(z)$ can be obtained about $z = -1$ in the regions (i) $0 < |z+1| < 1$

(ii) $1 < |z+1| < 3$ and (iii) $|z+1| > 3$.

27. Discuss the singularities of (i) $f(z) = \frac{e^z}{z^2 + 1}$ (ii) $f(z) = \frac{1 - \cos z}{z}$ (iii) $f(z) = e^{1/z}$

(iv) $f(z) = \frac{z^2}{(z-1)^2(z+2)}$.

Answer: (i) $z = -i, i$ are isolated singular points of $f(z) = \frac{e^z}{z^2 + 1}$

(ii) $z = 0$ is a removable singularity of $f(z) = \frac{1 - \cos z}{z}$

(iii) $z = 0$ is an essential singularity of $f(z) = e^{1/z}$

(iv) $z = 1, -2$ are the singular points of $f(z)$. Here $z = 1$ is a pole of order 2 and $z = -2$ is a simple pole.

28. Find the poles and the corresponding residues at each pole of the function

$$f(z) = \frac{z^2 + 2z}{(z+1)^2(z^2 + 4)}.$$

Answer: Here $z = -1$ is a pole of order two and $z = -2i, z = 2i$ are simple poles. And the

corresponding Residues are $\frac{2}{25}, \frac{1+7i}{2}, \frac{1-7i}{2}$ respectively.

29. Find the poles and the corresponding residues of the function $f(z) = \frac{z^2}{(z-1)(z-2)^2}$

Answer: Here $z = 1$ is a pole and $z = 2$ is a pole of order 2. And the corresponding Residues are 1, 0 respectively.

30. Evaluate $\int_C \frac{z+4}{z^2+2z+5} dz$ where C is the circle $|z| = 1$.

Answer: Clearly $f(z) = \frac{z+4}{z^2+2z+5}$ is analytic at all the points within and on the circle $|z| = 1$.

Therefore, by Cauchy's theorem, we have $\int_C \frac{z+4}{z^2+2z+5} dz = 0$.

31. Evaluate $\int_C \frac{z^3 e^{-z}}{(z-1)^2} dz$ where C is $|z-1| = \frac{1}{2}$.

Answer: By Cauchy's integral formula, $\int_C \frac{z^3 e^{-z}}{(z-1)^2} dz = \frac{\pi i}{e}$

32. Evaluate $\int_C \frac{z^2-1}{z^2+1} dz$ where C is $|z-i| = 1$.

Answer: By Cauchy's integral formula, $\int_C \frac{z^2-1}{z^2+1} dz = -2\pi$.

33. Evaluate $\int_C \frac{z^2+2z}{(z+1)^2(z^2+4)} dz$ where C is $|z| = \frac{3}{2}$ using Residue theorem.

Answer: Here $z = -1$ is a pole of order two which is inside of C and $z = -2i, z = 2i$ are simple

poles which are outside of C. Therefore $\int_C \frac{z^2+2z}{(z+1)^2(z^2+4)} dz = 2\pi i (\text{Res. } f(z))_{z=-1} = 2\pi i \left(\frac{2}{25} \right)$

34. Evaluate $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$ where C is $|z| = \frac{3}{2}$ using Residue theorem.

Answer: $\int_C \frac{4-3z}{z(z-1)(z-2)} dz = 2\pi i$.

35. Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos \theta} d\theta$ using Residue theorem.

Answer: $\frac{\pi}{6}$

36. Evaluate $\int_0^{\infty} \frac{2x^2-1}{x^4+5x^2+4} dx$ using Residue theorem.

Answer: $\frac{\pi}{4}$

37. Evaluate $\int_0^{\infty} \frac{x \sin x}{1+x^2} dx$ using Residue theorem.

Answer: $\frac{\pi i}{2e}$

38. Evaluate $\int_{-\infty}^{\infty} \frac{\cos 2x}{x} dx$.

Answer: 0